5.

a. $\Pi_{cid}(Customer) - \Pi_{cid}(\sigma_{ptype='chair'}(Product) \bowtie Purchase)$

b. $\Pi_{cid,name}(Customer) - \Pi_{cid,name}(Customer \bowtie Purchase)$

c. $Bought \leftarrow \Pi_{cid,pid}(Purchase \bowtie \sigma_{ptype="chair"}(Product))$

$BoughtTwo \leftarrow \Pi_{P1.cid}(\sigma_{P1.cid=P2.cid \wedge P1.pid \neq P2.pid}(\rho_{P1}(Bought) \times \rho_{P2}(Bought)))$

$BoughtOne \leftarrow \Pi_{cid}(Bought) - BoughtTwo$

$\therefore Result \leftarrow \Pi_{name}(Customer \bowtie BoughtOne)$

6.

a. R.A.: $\Pi_{flno}(\sigma_{to='HK'}(Flights))$

SQL:
```
SELECT flno
FROM Flights
WHERE to = 'HK';
```

b. R.A.: $\Pi_{aid}(Aircraft) - \Pi_{A1.aid}(\sigma_{A1.cruisingrange<A2.cruisingrange}(\rho_{A1}(Aircraft) \times \rho_{A2}(Aircraft)))$

SQL:
```
SELECT aid
FROM Aircraft
WHERE cruisingrange = (SELECT MAX(cruisingrange) FROM Aircraft);
```

C. R.A.: $\Pi_{eid}(Certified \bowtie \sigma_{cruisingrange>=8000}(Aircraft))$

SQL:
```
SELECT DISDINCT C.eid
FROM Cetified C
JOIN Aircraft A ON C.aid = A.aid
WHERE A.cruisingrange >= 8000;
```

d. R.A.: $Direct \leftarrow \Pi_{aid}(\sigma_{from='New York' \wedge to='Los Angeles' \wedge distance<=cruising}(Flights \times Aircraft))$

$Pilots \leftarrow \Pi_{eid}(Certified \bowtie Direct) - \Pi_{eid}(Certified \bowtie \sigma_{amanufacturer="Airbus"}(Aircraft))$

$Result \leftarrow \Pi_{ename}(Employees \bowtie Pilots)$

SQL:
```
SELECT DISTINCT E.ename
FROM Employees E
JOIN Certifred C ON E.eid = C.cid
JOIN Aircrafts A ON C.aid = A.aid
JOIN Flights F ON F.distance <= A.cruisingrange
WHERE F.from = 'New York'
   AND F.to = 'Los Angeles'
   AND E.eid NOT IN(
        SELECT C2.eid
        FROM Certified C2
        JOIN Aircraft A2 ON C2.aid = A2.aid
        WHERE A2.amanufacturer = 'Airbus'
        );
```

e. R.A.: $Boeing \leftarrow \Pi_{aid}(\sigma_{amanufacturer='Boeing'}(Aircraft))$

$CertifiedBoeing \leftarrow Certified \bowtie Boeing$

$EidCount \leftarrow GroupBy\ eid(Count(aid)\ from\ CertifiedBoeing)$

$Result \leftarrow \Pi_{Salary}(\Pi_{eid}(\sigma_{count=3}(EidCount)) \bowtie Employees)$

SQL: see next page.

SQL:
```
SELECT E.salary
FROM Employees E
JOIN Certified C ON E.eid = C.cid
JOIN Aircraft A ON C.aid = A.aid
WHERE A.amanufacturer = 'Boeing'
GROUP By E.eid, E.salary
HAVING COUNT(A.aid) = 3;
```

here "s" means there may be more than one employees having the same highest salary

7. This expression is querying for the (ID, person_name)s of employees with the highest salary